

From Random Packing to an Interpretable Transport Closure: Structure-Resolved CO₂ Adsorption in DEM-Generated Zeolite Beds

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Abstract

Packed beds with almost the same porosity can contain different connected void pathways. These differences may change how quickly carbon dioxide reaches and uses the adsorbent, but they are hidden by conventional bulk descriptors. We study this effect without computational fluid dynamics. Twenty beds containing 1850 spherical commercial zeolite 13X beads were generated by the discrete- element method. Their void spaces were converted into pore networks, on which a conservative diffusion-adsorption model was solved. Gas diffusion through pore throats was coupled to one linear-driving-force adsorption equation per bead. Only the random packing realization changed between cases. The robust bulk porosity varied by about 0.55%, whereas CO₂ uptake at 5000 s varied by 11.0%. A homogeneous one-dimensional model reproduced the network uptake curves with a mean time-weighted normalized error of 1.58%, but its fitted effective diffusivity varied by 23.1%. Porosity could not predict this coefficient in leave-one-bed-out tests ($R^2 = -0.234$). In contrast, an inlet-to-interior pore-network conductance produced an accessibility diffusivity that correlated strongly with the fitted coefficient. A positive power-law closure predicted withheld beds with $R^2 = 0.890$. The results show that connectivity and entrance accessibility contain transport information that is absent from bulk porosity. The framework provides an interpretable bridge from particle-resolved geometry to a cheap homogeneous adsorption model, while keeping adsorption thermodynamics and mass conservation explicit.

Keywords: CO₂ capture; zeolite 13X; packed bed; DEM; pore network; adsorption; effective diffusivity; physical closure

1 Introduction

Packed beds of porous particles are widely used for gas separation and CO₂ capture. Their performance depends on two different scales. At the particle scale, the adsorbent material determines equilibrium capacity and uptake rate. At the bed scale, the arrangement of particles determines the empty pathways through which gas reaches the adsorbent. Standard one-dimensional models usually represent this second scale using porosity and one effective transport coefficient. This is efficient, but porosity alone does not describe whether the void space is well connected or contains entrance bottlenecks.

Particle-resolved computational fluid dynamics could examine these pathways, but it is expensive for an ensemble of stochastic beds. Here we use a simpler physics-based route. The discrete-element method (DEM) creates mechanically realistic particle arrangements. A pore network extracted from each packing then represents the void space as connected pore bodies and throats [1]. CO₂ diffusion is solved on this network and coupled to adsorption by every zeolite bead.

The goal is not to reproduce a flowing industrial adsorption column. The bed contains stagnant gas, and CO₂ enters by diffusion from a reservoir at the bottom. This restricted problem isolates

the effect of packing structure. It allows us to ask a precise question: can a simple, measurable property of the pore network explain packing-to-packing changes that bulk porosity misses?

The main contribution is an interpretable closure linking a geometry-derived accessibility conductance to the effective diffusivity of a homogeneous one-dimensional adsorption model. The study deliberately uses simple physical descriptors rather than a graph neural network because only twenty independent packings are presently available.

2 Physical problem and modelling workflow

The bed initially contains nitrogen with zero CO_2 concentration and the zeolite beads initially contain no adsorbed CO_2 . The lower boundary is connected to a reservoir at $c_{\text{in}} = 6.05 \text{ mol m}^{-3}$. The top and cylindrical wall are closed. Temperature is constant at 298.15 K. There is no imposed velocity, pressure-drop calculation, heat of adsorption or CFD.

Figure 1 illustrates the model. Large circles represent zeolite beads. Small blue nodes and connecting lines represent the pore network. Gas diffuses through this network and is adsorbed by adjacent particles.

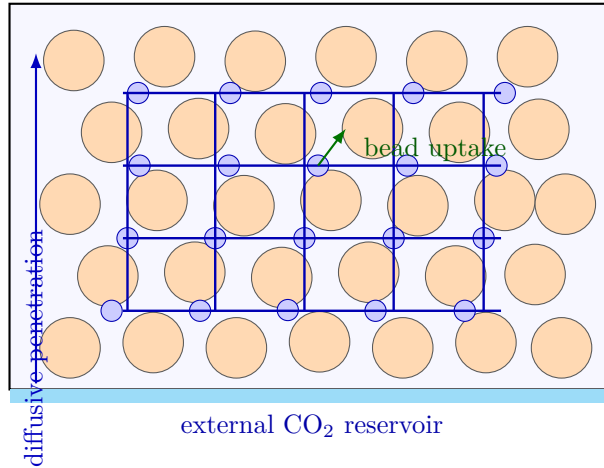


Figure 1: Schematic of the diffusion–adsorption problem. DEM supplies the particle positions. A geometry-derived pore graph carries gas-phase CO_2 , and each particle has its own adsorbed loading. This vector schematic is drawn directly in this TeX file and needs no external data.

The complete workflow is shown in Fig. 2. DEM and the pore network provide the detailed reference model. The homogeneous model is the cheap reduced description. The closure connects them.

3 Methods

3.1 DEM packing ensemble

Twenty independent cylindrical beds were generated with LAMMPS. Every bed contains 1850 monodisperse spheres of diameter 1.6 mm and apparent density 1600 kg m^{-3} . The tube diameter is 16 mm, giving $D_{\text{tube}}/d_p = 10$. Gravity, Hertz–Mindlin contacts, friction and damping were used to fill and settle the beds. Only the insertion seed changed. A packing was accepted only if it contained all particles, was mechanically settled and passed overlap and geometry checks.

Bulk porosity was calculated using a robust bed top defined by the 99th percentile of particle-top elevations. Sphere intersections with axial and radial bins were integrated instead of assigning each complete sphere to one bin. This prevents artificial jumps in local porosity.

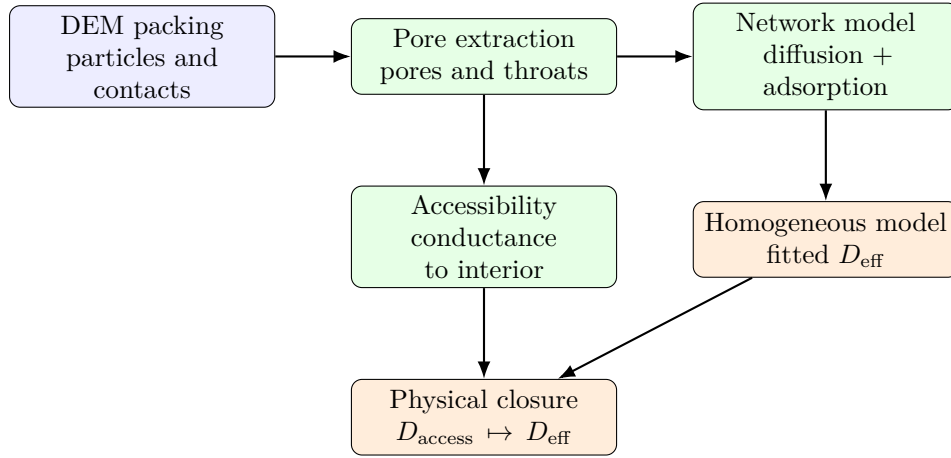


Figure 2: Physics-focused hybrid workflow. The pore network generates detailed, conservative adsorption curves, while the homogeneous model provides the cheap description used at larger scales. This vector diagram is contained in the TeX source.

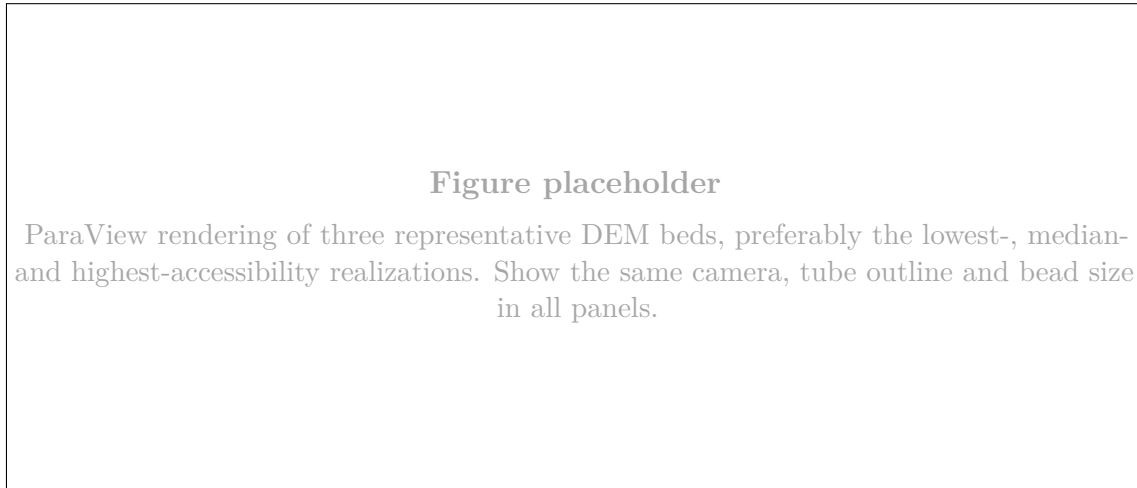


Figure 3: Representative DEM-generated zeolite beds. The final figure should show that all cases have the same particle and tube sizes but differ slightly in local arrangement.

Figure source: Open the particle VTP files in `paraview_co2_packings/seed_*/particles_final.vtp`. If seed subfolders were not retained, regenerate them from `co2_13x_seed_*/particles_final.dump` using `convert_co2_packings_to_vtp.py`.

3.2 Pore-network extraction

The void geometry was converted to a Delaunay–Voronoi pore network. A pore body stores gas volume and concentration. Two pore bodies are connected by a throat when their Delaunay tetrahedra share a face. For a throat connecting pores i and j , the diffusive conductance is

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad (1)$$

where $D_m = 1.5 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$, A_{ij} is the open throat area and L_{ij} is its length. Pore volumes were allocated using $2^{22} = 4,194,304$ Sobol samples. All networks were connected, spanned the bed and conserved total void volume. For example, seed 18427 contained 9397 pores and 18104 throats.

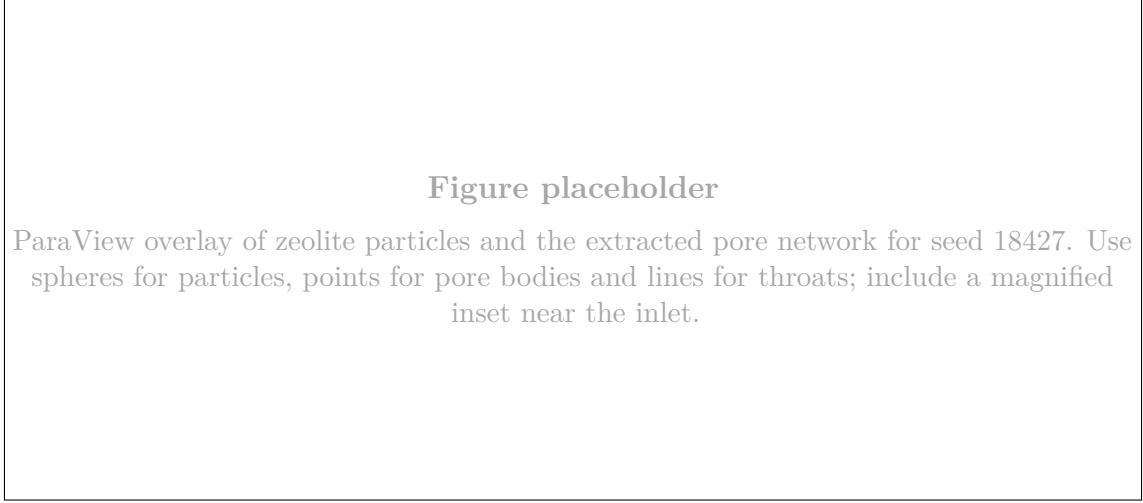


Figure 4: Particle packing and corresponding pore network for the verification case. The overlay provides a direct geometric check that pore nodes occupy the void space and throats connect neighboring void regions.

Figure source: Use `paraview_co2_packings/.../particles_final.vtp` together with `co2_pore_networks_power22/seed_18427/pore_network.vtp`. The underlying tables are `pores.csv` and `throats.csv` in the same network directory.

3.3 Conservative diffusion–adsorption model

Each pore i has gas concentration c_i and volume V_i . Each zeolite bead k has mean adsorbed loading q_k in mol kg^{-1} . The gas balance is

$$V_i \frac{dc_i}{dt} = \sum_j G_{ij}(c_j - c_i) - \sum_k w_{ki} m_k \frac{dq_k}{dt}. \quad (2)$$

The weights w_{ki} distribute bead uptake among adjacent pores and sum to one for each bead. Therefore, every mole removed from the gas is added to the solid.

Equilibrium loading is represented by the Langmuir form

$$q_k^* = \frac{q_s b p_k}{1 + b p_k}, \quad p_k = c_k^g R T, \quad (3)$$

and bead uptake follows the linear-driving-force equation

$$\frac{dq_k}{dt} = k_{\text{LDF}}(q_k^* - q_k). \quad (4)$$

The adopted parameters are $q_s = 5.332 \text{ mol kg}^{-1}$, $b = 5.093 \times 10^{-5} \text{ Pa}^{-1}$ and $k_{\text{LDF}} = 8.55 \times 10^{-3} \text{ s}^{-1}$. The affinity assumes that a published table omitted a factor of 10^{-3} ; this assumption is discussed in Sec. 6. The kinetic coefficient is inferred from the commercial 13X bead time constant reported by Hu et al. [3].

At the inlet, every labelled inlet pore is connected to the same external reservoir. The nominal areas assigned to these connections sum to the full tube cross-section, $A_{\text{tube}} = \pi R^2$. Internal blockage remains represented by the pore and throat geometry. The stiff coupled equations were integrated with an implicit BDF method and an analytical sparse Jacobian.

3.4 Accessibility descriptor

The main structural descriptor is a steady conductance from the inlet to a plane $5d_p$ inside the bed. Inlet-pore concentration is set to one and the interior-plane concentration to zero. Conservation is solved in all pores between them. The resulting total flux is the effective accessibility conductance G_{access} . It is converted to diffusivity units by

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}. \quad (5)$$

The same calculation was repeated at $3d_p$, $7d_p$ and $10d_p$ to check that the conclusion did not depend on one analysis plane.

3.5 Homogeneous one-dimensional model

The reduced model divides the bed height into 100 finite-volume cells. It does not contain individual pores or particles:

$$\varepsilon_b \frac{\partial c}{\partial t} = D_{\text{eff}} \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t}, \quad (6)$$

$$\frac{\partial q}{\partial t} = k_{\text{LDF}} [q^*(c) - q]. \quad (7)$$

The isotherm, kinetic coefficient, inlet concentration, bed height and initial conditions are the same as in the network model. Only D_{eff} is fitted for each bed. The inlet uses a Robin condition that includes both the external resistance and diffusion across half of the first numerical cell. The fit minimizes a time-integrated squared uptake error.

3.6 Statistical analysis and closure

Predictive tests leave out one complete bed, fit the model to the remaining 19 beds and predict the omitted bed. This is repeated twenty times. We report the resulting R^2 , RMSE and mean absolute percentage error. Parameter uncertainty is estimated by 10,000 bed-level bootstrap samples. Target permutation tests use 5000 random permutations.

Three useful closure forms are compared:

$$D_{\text{eff}} = a_0 + a_1 \varepsilon_b, \quad (8)$$

$$D_{\text{eff}} = \alpha D_{\text{access}}, \quad (9)$$

$$D_{\text{eff}} = C D_{\text{access}}^n. \quad (10)$$

Affine conductance models are also evaluated as numerical baselines. The power law is reported in normalized form so that its coefficient is dimensionless.

4 Results

4.1 Matched porosity but different adsorption response

All twenty packings passed mechanical and pore-network quality checks. Mean robust bulk porosity was 0.44665 and its coefficient of variation was about 0.55%. The mean bed height was 35.61 mm. Despite this close global matching, uptake at 5000 s ranged from 0.2056 to 0.3148 mol kg⁻¹, with a coefficient of variation of 11.04%. The mean relative half-uptake time was 1501 s.

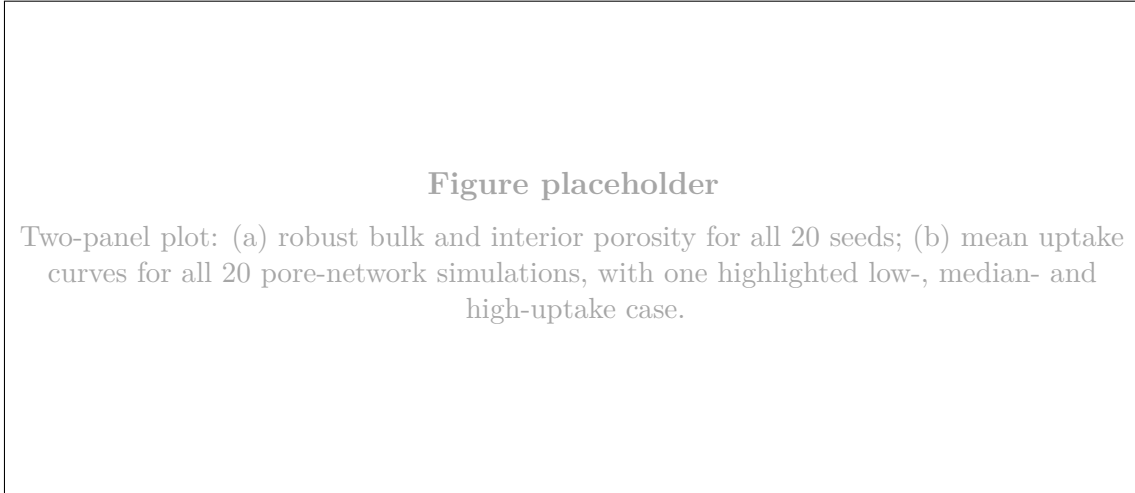


Figure 5: Global packing similarity and transient response variability. Panel (a) should demonstrate the narrow porosity range, whereas panel (b) should show that the adsorption curves remain meaningfully different.

Figure source: Porosity data and existing profiles are in `co2_porosity_analysis/`. Uptake curves are in `co2_adsorption_campaign_finite_inlet_5000s/seed_*/timeseries.csv`; campaign values are in `co2_adsorption_finite_inlet_analysis/co2_adsorption_campaign_dataset.csv`.

4.2 Spatially heterogeneous utilization

The bed remained far from equilibrium at 5000 s. For seed 18427, mean loading was 0.2843 mol kg⁻¹, only about 12.3% of the equilibrium loading at the inlet concentration. CO₂ entered from below, so lower particles were used more strongly than particles deeper in the bed. Particle-loading coefficients of variation ranged from 1.79 to 2.03 across the ensemble.

4.3 Accessibility explains transient utilization

The inlet-to-interior conductance correlates with uptake at 5000 s with Spearman $\rho = 0.968$. It correlates negatively with the underutilized-particle fraction ($\rho = -0.959$). The unique number of second-layer pores correlates with half-uptake time ($\rho = -0.814$). Porosity-only leave-one-bed-out models give negative R^2 values for these responses. Conductance alone predicts uptake with $R^2 = 0.924$ and the underutilized fraction with $R^2 = 0.863$.

4.4 Homogeneous reduction

The 100-cell homogeneous model fitted all twenty network curves successfully. The mean time-weighted normalized RMSE was 1.58%, with a range from 0.84 to 2.46%. Fitted D_{eff} ranged from 4.340×10^{-7} to 1.096×10^{-6} m² s⁻¹, giving a coefficient of variation of 23.13%. Thus,

Figure placeholder

ParaView particle-loading maps at selected times, for example 100, 1000 and 5000 s, or a final comparison of low- and high-accessibility beds. Use the same color scale in every panel.

Figure 6: Spatial development of particle loading. A common color range is essential so that penetration and underutilized regions can be compared directly.

Figure source: Particle coordinates are in `co2_13x_seed_*/particles_final.dump`. Particle loadings are in `co2_adsorption_campaign_finite_inlet_5000s/seed_*/final_state.npz`. Use the pore-network `particle_pore_incidence.csv` if mapping support is needed. A publication-figure script can write the combined VTP files.

Figure placeholder

Three scatter plots with seed labels or selected labels: (a) porosity versus uptake, (b) inlet-to-interior conductance versus uptake, and (c) conductance versus underutilized-particle fraction. Include bootstrap trend intervals and report Spearman correlation in each panel.

Figure 7: Why bulk porosity is insufficient. The figure should contrast the absence of a useful porosity trend with the strong accessibility relationships.

Figure source: Use `co2_adsorption_finite_inlet_analysis/co2_adsorption_campaign_dataset.csv`, the merged structure table produced by `extract_co2_inlet_accessibility.py`, and statistical outputs in `co2_reduced_physical_model_analysis/`.

the homogeneous model is accurate, but its transport coefficient must retain packing-specific information.

A 100–200–400-cell comparison for seed 18427 changed D_{eff} by 0.156% and then 0.039%, supporting the 100-cell choice. The maximum absolute mass-balance residual in the accepted test was 1.38×10^{-13} mol.

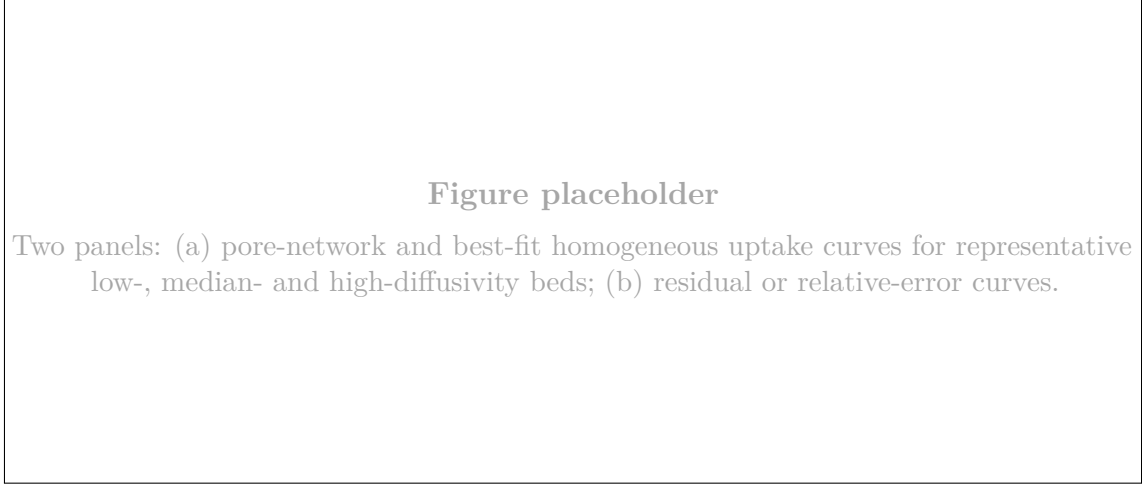


Figure 8: Accuracy of the one-dimensional homogeneous reduction. Curves should be evaluated over the full 0–5000 s interval, not only at the final time.

Figure source: Network curves are in `co2_adsorption_campaign_finite_inlet_5000s/seed_*/timeseries.csv`. Homogeneous curves and fit comparisons are in `co2_homogeneous_campaign_100cells/seed_*/timeseries.csv` and `fit_comparison.csv`; campaign parameters are in `co2_homogeneous_campaign_100cells/homogeneous_c`

4.5 Effective-diffusivity closure

At $5d_p$, D_{access} and fitted D_{eff} have Pearson correlation 0.951 and Spearman correlation 0.953. A porosity-only model gives leave-one-bed-out $R^2 = -0.234$. A direct proportional closure, $D_{\text{eff}} = \alpha D_{\text{access}}$, gives $\alpha = 0.1523$ with bootstrap interval 0.1427–0.1616, but its predictive R^2 is only 0.555. It is therefore too restrictive.

The affine accessibility model reaches $R^2 = 0.881$, but its negative intercept can produce negative diffusivity outside the fitted range. The positive power law gives almost the same predictive accuracy without this problem:

$$\frac{D_{\text{eff}}}{D_m} = 0.7589 \left(\frac{D_{\text{access}}}{D_m} \right)^{2.4969}. \quad (11)$$

Its leave-one-bed-out R^2 is 0.890. The bootstrap median exponent is 2.504 with a 95% interval of 2.057–2.768. The raw-conductance affine model gives a slightly higher $R^2 = 0.898$, but also has a negative intercept and is less portable because it does not account for bed height and area.

4.6 Sensitivity to the interior-plane depth

Changing the interior plane from $3d_p$ to $10d_p$ reduces conductance because the diffusion path becomes longer. Nevertheless, Pearson correlations with D_{eff} remain between 0.936 and 0.951. Affine leave-one-bed-out R^2 remains between 0.834 and 0.881. The result is therefore not caused by the arbitrary choice of one plane. The $5d_p$ definition is retained because it lies beyond the immediate inlet layer and gives the strongest prediction.

Figure placeholder

Two panels: (a) fitted effective diffusivity versus accessibility diffusivity, showing the positive power-law curve and bootstrap band; (b) observed versus leave-one-bed-out predicted diffusivity for the porosity, proportional and power-law models.

Figure 9: Formal closure from pore-network accessibility to homogeneous effective diffusivity. The one-to-one line in panel (b) should be shown.

Figure source: All required joined values are in `co2_effective_diffusivity_closure/closure_dataset.csv`. Use `closure_loo_predictions.csv`, `closure_bootstrap_intervals.csv` and `closure_summary.json` in the same directory. Preliminary plots `closure_physical_fit.*` and `closure_loo_predictions.*` are also written there.

Figure placeholder

Depth-sensitivity plot showing correlation and leave-one-bed-out R-squared at 3, 5, 7 and 10 particle diameters. A second axis or panel may show the decrease in mean conductance with depth.

Figure 10: Robustness to the definition of the inlet-accessibility depth.

Figure source: Use `co2_effective_diffusivity_closure/closure_depth_sensitivity.csv` and the four input files `co2_inlet_accessibility_{3,5,7,10}dp.csv`.

5 Discussion

The results separate three ideas that are often combined in one effective parameter. First, porosity measures how much empty volume exists. Second, connectivity determines whether this volume forms useful transport paths. Third, entrance accessibility determines how easily CO₂ reaches those paths from the boundary. The twenty beds have almost the same amount of void space but different values of the latter two properties.

The homogeneous model shows that the detailed network curves can be reduced to one packing-specific transport coefficient with small error. This is useful: the pore network is needed to understand and construct the closure, but it does not need to be solved whenever a larger-scale model uses the bed. Equation (11) transfers the relevant geometric information into the cheap one-dimensional model.

The power-law exponent is greater than one. Within the studied range, this means that small improvements in inlet-to-interior accessibility accompany larger relative improvements in fitted bed transport. This should be treated as an empirical structural result, not yet as a universal law. The twenty beds share one tube-to-particle ratio, bead size and material parameter set.

The study also illustrates why the statistically best equation is not always the best physical closure. The raw-conductance affine model has a slightly higher cross-validated R^2 , but its negative intercept eventually predicts negative diffusivity. The selected power law is always positive, is naturally normalized and sacrifices little predictive accuracy.

6 Limitations

The model describes dry, isothermal, diffusion-driven filling from one side. It does not describe pressure-driven breakthrough, pressure drop, heat release, water competition or regeneration. The conclusions therefore concern the influence of packing geometry on transient accessibility, not complete industrial capture performance.

The adopted zeolite affinity $b = 5.093 \times 10^{-5} \text{ Pa}^{-1}$ assumes that the source table omitted a factor of 10^{-3} . The LDF coefficient is converted from a reported single-bead time constant at a somewhat higher temperature [3]. These values provide a documented and internally consistent working case, but new material measurements would be needed for quantitative prediction of a particular commercial batch.

Finally, all statistical observations come from twenty realizations of one geometry. Leave-one-bed-out validation tests interpolation among these beds; it does not test transfer to other particle sizes or tube-to-particle ratios. Such external validation is a valuable later step but is outside the completed scope reported here.

7 Conclusions

Twenty DEM-generated zeolite beds with closely matched porosity produced meaningfully different transient CO₂ adsorption responses. A conservative pore-network model traced these differences to inlet accessibility and internal connectivity. A homogeneous one-dimensional model reproduced the detailed uptake curves with a mean normalized error of 1.58%, but required effective diffusivities that varied by 23.1% between beds.

Porosity could not predict this variation. A dimensionless positive power law based on inlet-to-interior accessibility predicted withheld-bed effective diffusivity with $R^2 = 0.890$. The work therefore provides a simple physical message: equal void volume does not imply equal transport. How that void space is connected to the inlet matters. The proposed closure preserves this information in a form that can be used by a cheap homogeneous adsorption model without CFD or a high-capacity machine-learning model.

Data and code availability

The final publication package should contain the LAMMPS input, prescribed seed list, packing-quality summary, pore-network extraction and QA scripts, diffusion-adsorption solver, homogeneous fitting scripts, closure-analysis script and the processed CSV tables identified in the figure captions. A public repository DOI and archive link will be inserted here before submission.

References

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